

Week 1.2 — Soldering Bootcamp

Friday, June 19, 2026 (Sunday cohort upcoming June 21) · Two sessions · Friday cohort documented; Sunday pending

Lesson 1.2 was soldering bootcamp — the team's first night with hot irons in their hands. Before the rest of the team touched the kits, Coach John ran Trenton — who had prior soldering experience — through a full test build on Friday June 18, the day before the live class. That prep session is where the four rules below actually came from. Use an experienced student to surface lessons before the new ones make them: that's the mentor pattern, and it kept Friday's class clean. We split the live class across Friday June 19 and Sunday June 21 because we had more kids than workstations. Friday's cohort — Rebecca, Isaac, Liam, Maddox, and Tyler — ran the full sequence: safety brief, general soldering instruction, then practice board. Coach John ran the brief and led the instruction. Each kid built the SpaceWings DIY Lucky Wheel — an NE555 + CD4017 spinner that lights ten LEDs in sequence. Through-hole, mixed small and big components, immediate visual payoff when you press the button. About an hour per board. Iron set to 650°F, which flows good solder cleanly without cooking the components. Sunday's cohort meets next; the rest of the team builds the same kit then.

2

SESSIONS

5/12

STUDENTS FRIDAY

~1 hr

PER BOARD

650°F

IRON TEMP

SAFETY BRIEF

- Solder is lead-based. Don't inhale the fumes; wash hands when you're done.
- Place a fan to extract or blow fumes away from the student. Direction matters — away, not toward.
- The iron tip burns. So do the back of the board and the component leads. Treat both as hot.
- Safety glasses on whenever you're snipping leads. Small metal clippings fly off the cutters.

DECISIONS MADE

- Coach John runs the safety brief and the general soldering instruction at each session — the standing pattern.
- Trenton handles pre-class test builds on new kits, to surface lessons before the rest of the team touches an iron.
- Iron temperature standardized at 650°F for the team's practice kits.
- Split the class across two sessions when workstations are the bottleneck.
- Safety brief at the start of every soldering session. No exceptions.

ACTION ITEMS

ITEM	OWNER	DUE
Run the Sunday cohort through the same kit + brief + instruction sequence	Coach John	Sun 6/21
Bring kits and components to next class (Lesson 1.3)	Advisor	6/28
Practice soldering at home if a station is available	Every student	ongoing
Add team photos from Friday + Sunday sessions to the photo log once parent releases are returned	Photographer	when consent is in

WHAT WE LEARNED

- Work from the inside of the board outward. Anything you've already soldered is a vertical obstacle to whatever you solder next. Building outward keeps your iron's path open.
- Small components before big ones. Resistors and low-profile caps first; headers and tall connectors last. Same reasoning — don't build a wall you have to reach over.
- Right-handed solderers do multi-pin parts left to right. Left-handed, right to left. Your hand and iron drag across joints you just made if you go the other direction, and heat plus motion is exactly how a clean joint becomes a smeared one.
- Solder the battery pack last. The pack hangs off the PCB and blocks access to every joint underneath it. Mount it after everything else is on the board and your workspace stays clean to the end.

STUDENT OUTCOMES

- The importance of polarity in certain components — diodes, electrolytic caps, transistors — and the habit of slowing down to check it before you commit the joint.
- Soldering technique, equipment care, and the safety habits that go with hot tools and lead-based material.

LOOKING AHEAD

Sunday June 21 brings the rest of the team through the same kit + brief + instruction sequence. After that, Class 1.3 — Kit Unboxing and First Electronics — on June 28. Kits get opened, components get identified, and we move from practice joints to circuits that actually do something. Closed-toe shoes still required.

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